6 (a) (i) When the white light is incident on the gas, photons with wavelengths that correspond to the energy difference between any two energy levels are absorbed by the atoms. During de-excitation, these photons are emitted in all [B1] [B1]

directions. In the resulting spectrum, these wavelengths correspond to dark spectral lines on a coloured background. [B1]

(ii)

$$\Delta E_{12} = hf = h\frac{c}{\lambda_{12}}$$

$$\lambda_{12} = \frac{hc}{\Delta E_{12}}$$

$$= \frac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{\left[\left(-3.04\right)-\left(-5.14\right)\right]\times\left(1.60\times10^{-19}\right)}$$ [M1]

$$= 592 \text{ nm (shown)}$$ [A0]

(iii)

$$\lambda_{13} = \frac{hc}{\Delta E_{13}}$$

$$= \frac{\left(6.63\times10^{-34}\right)\left(3.00\times10^{8}\right)}{\left[\left(-1.53\right)-\left(-5.14\right)\right]\times\left(1.60\times10^{-19}\right)}$$

$$= 344 \text{ nm}$$ [M1]

Since wavelength of light corresponding to transition from E_1 to E_3 is already outside visible region, there is no need to test for transition from E_1 to E_4.

Hence, only one absorption spectral line, corresponding to the E_1 to E_2 transition, is observed. [A1]

OR OR

$$E_{max} = \frac{hc}{\lambda_{min}} = \frac{6.63\times10^{-34}\times3\times10^{8}}{400\times10^{-9}} = 4.97\times10^{-19} \text{ J} = 3.108 \text{ eV}$$

$$-5.14 + 3.108 = -2.03 \text{ eV} < -1.53 \text{ eV}$$ [M1]

Since only level E_2 is attainable from level E_1, only one absorption spectral line, corresponding to the E_1 to E_2 transition, is observed. [A1]

(b) (i)

$$\text{KE} = \frac{p^2}{2m} = \frac{(h/\lambda)^2}{2m}$$

$$= \frac{(6.63\times10^{-34} / 592\times10^{-9})^2}{2\times(9.11\times10^{-31})}$$ [M1]

$$= 6.88\times10^{-25} \text{ J}$$

$$= \frac{6.88\times10^{-25}}{1.6\times10^{-19}} = 4.30\times10^{-6} \text{ eV (3 s.f.)}$$ [A1]

(ii)

- Higher resolution / higher magnification B1
- due to shorter wavelengths used as energy of the electron can be increased by acceleration it to high speeds. B1

c	**i**	Probability density of finding the electron at distance r.	B1
	ii	Probability density is greatest near $r = 0$.	B1
		Therefore the electron is most likely to be detected close to the nucleus.	B1